Do Now · retrieval
Data mining and visualisation

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is data mining?

Q2

What is an outlier?

Q3

What type of graph shows two numeric variables?

Q4

C# int[] scores={2,4,6,8}. Double elements at indexes 0 and 2; give the final values and total.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Identify the question — Data mining searches for useful patterns in a set of records; decide what question the analysis should answer.

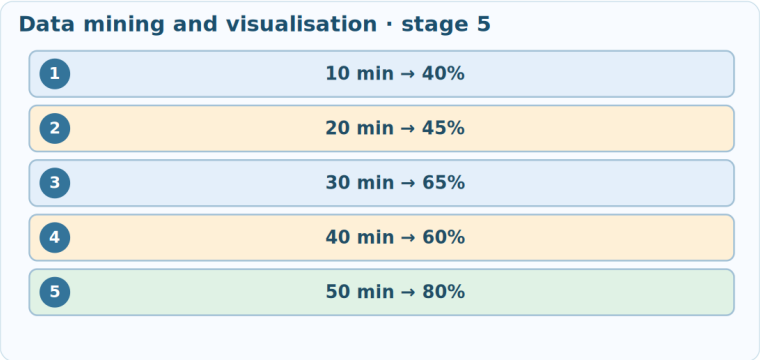
Core idea 2

Choose a suitable chart — Scatter charts show relationships between two numeric variables; bars compare categories; histograms show numeric distributions.

Core idea 3

Interpret responsibly — An association does not prove that one variable causes another. Consider sample size, bias, missing data and outliers.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

data mining — Extracting patterns from datasets.

visualisation — Graphical representation of data.

correlation — Statistical association, not proof of cause.

outlier — Observation unusually far from others.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

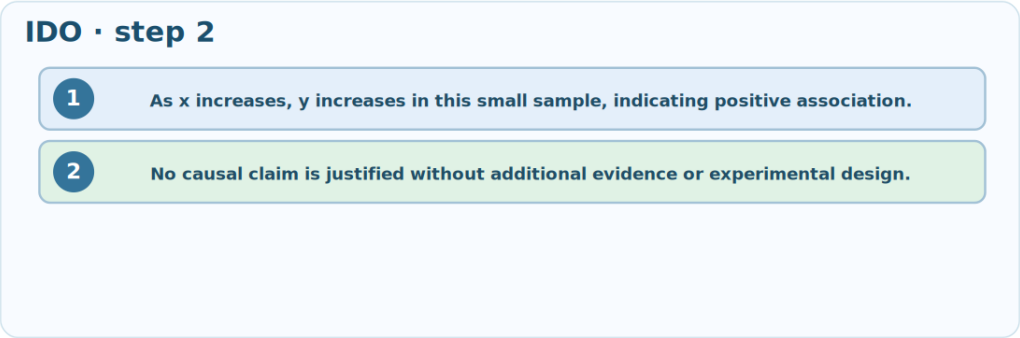
For observations (1,2), (2,4), (3,6), describe the association without claiming causation.

Follow the worked method

1. Predict first — As x increases, y increases in this small sample, indicating positive association.

2. Explain the method — No causal claim is justified without additional evidence or experimental design.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

A box plot or ordered dot plot shows the unusually high score 90 clearly.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Choose and justify a chart for (a) scores by category (b) minutes versus score (c) frequency by score interval. [6]

Question 2

For [40,42,45,90], calculate mean, median and identify one misleading interpretation. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate a school's plan to mine pupil browsing records to predict examination outcomes. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

If two variables are correlated, the first must cause the second.

Use key vocabulary from memory

1. Explain data mining in your own words.

2. Explain visualisation in your own words.

3. Explain correlation in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.